Digital Forensics Investigation: Suspected Intellectual Property Theft

- This project simulates the examination of a forensic disk image belonging to a former employee of an engineering firm suspected of violating company policies and accessing restricted intellectual property.
  - **Disclaimer:** This project was conducted in a controlled educational environment for cybersecurity training and portfolio development purposes. No production systems were accessed or targeted.
- **Method:** The investigation was performed using the Autopsy Digital Forensics Platform and standard forensic examination procedures.

Objectives

- Conduct a forensic examination of the provided disk image using forensic best practices.
- Identify evidence of unauthorized access to confidential corporate information that could support allegations of wrongdoing by the employee.
- Recover deleted files and artifacts.
- Evaluate evidence relevant to intellectual property theft allegations.
- Document findings and provide an investigative conclusion.

Scope of Investigation

- **Scenario:** Johnny Smithers was employed as an accountant for an engineering firm. He signed a Non-Disclosure Agreement (NDA) and an Acceptable Use Policy (AUP). He is suspected accessing and retaining company Intellectual Property (IP) that he was not authorized to access.

  Fundamental Question: Is there any digital evidence that Mr. Smithers accessed IP without authorization?

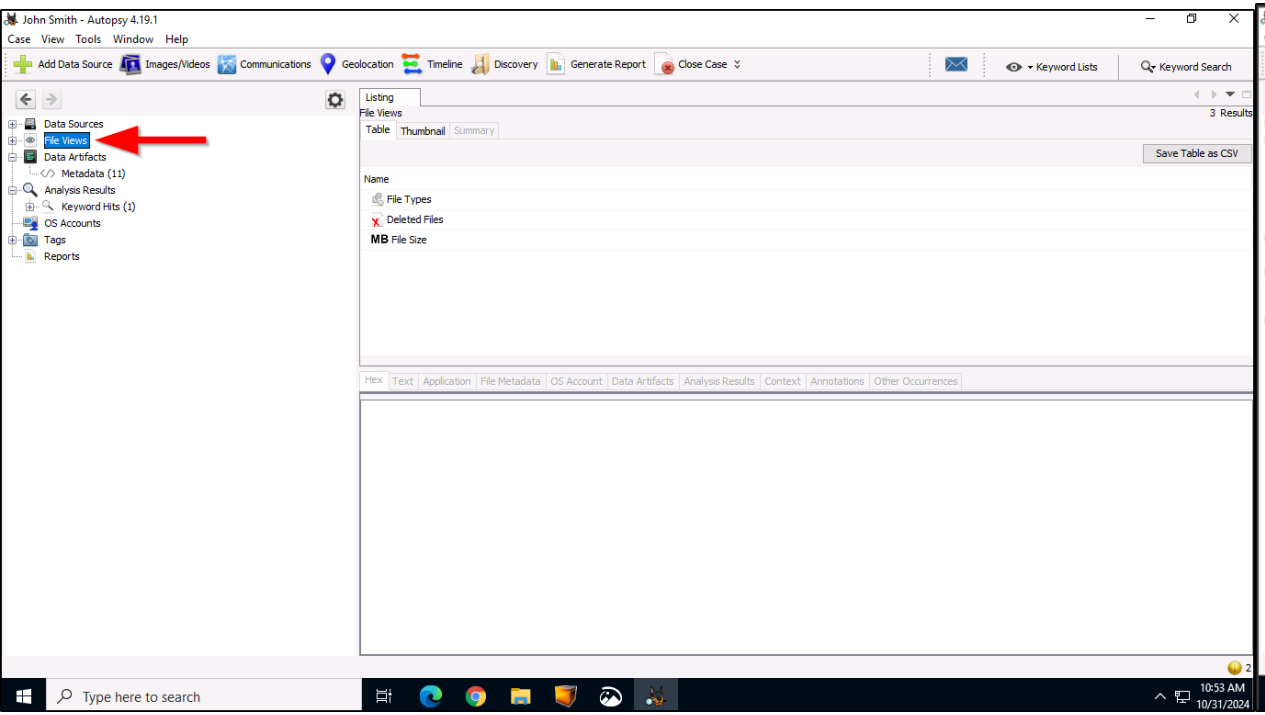
- **Method:** I will examine a forensic copy of Mr. Smithers' storage drive via Autopsy to ascertain if there is digital evidence to support the allegations.

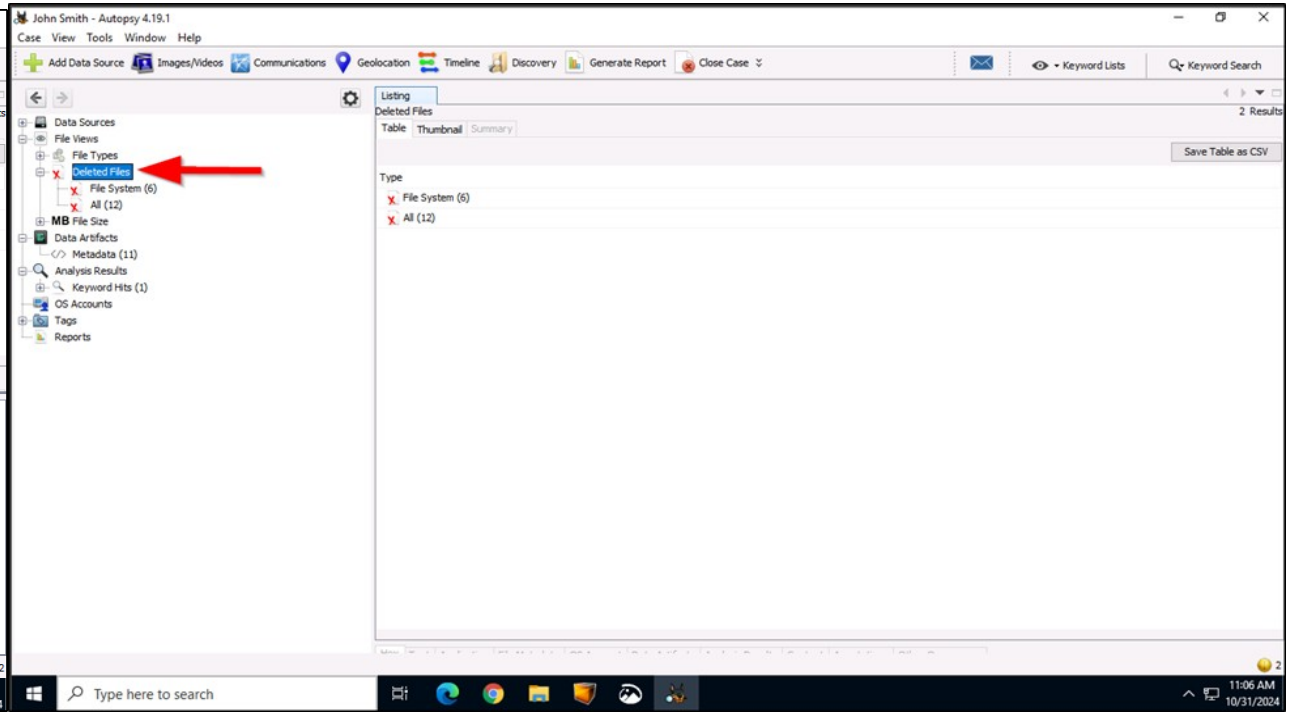
Tools and Technology Used

- **Forensic Software**
  - Autopsy Digital Forensics Platform
- **Investigative Techniques**
  - Evidence Preservation
  - Disk Image Analysis
  - Deleted File Recovery
  - Artifact Examination
  - Document Classification
  - Digital Evidence Bookmarking
  - Investigative Reporting
- **Cybersecurity Domains**
  - Digital Forensics
  - Incident Response
  - Insider Threat Investigation
  - Intellectual Property Protection
  - Evidence Analysis

Phase 1: Evidence Acquisition

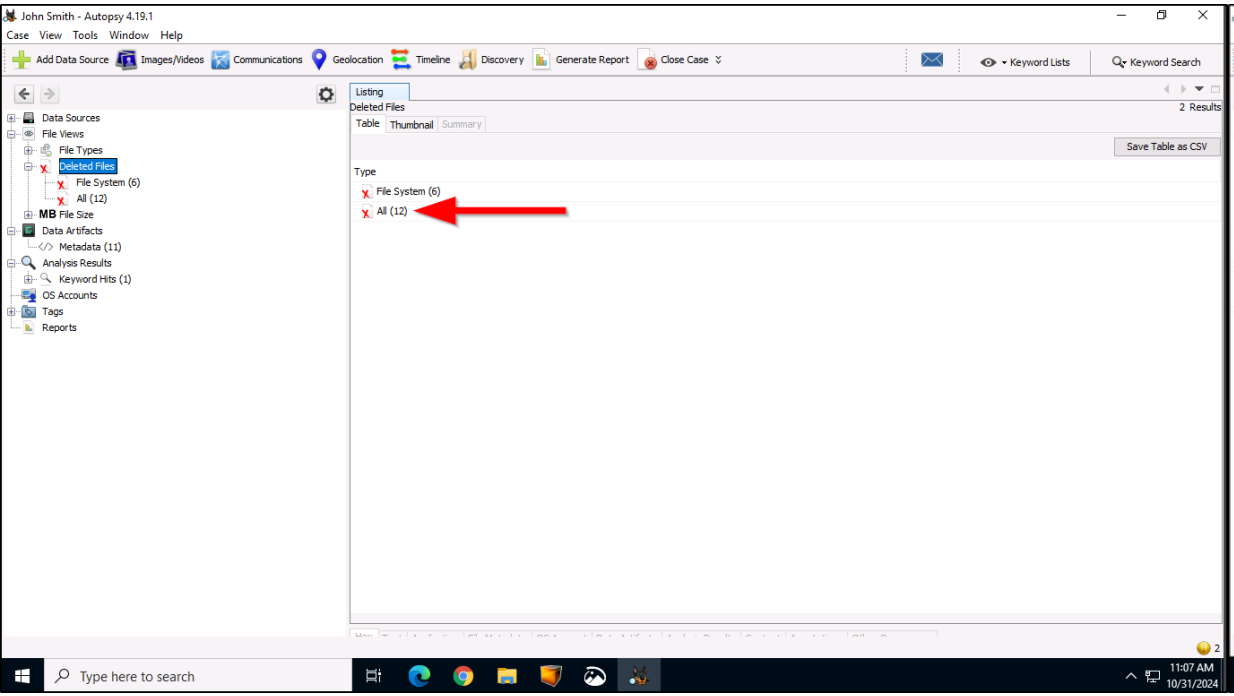
A forensic disk image was imported into Autopsy and processed using standard forensic analysis modules. The investigation was conducted against the forensic copy to preserve evidence.

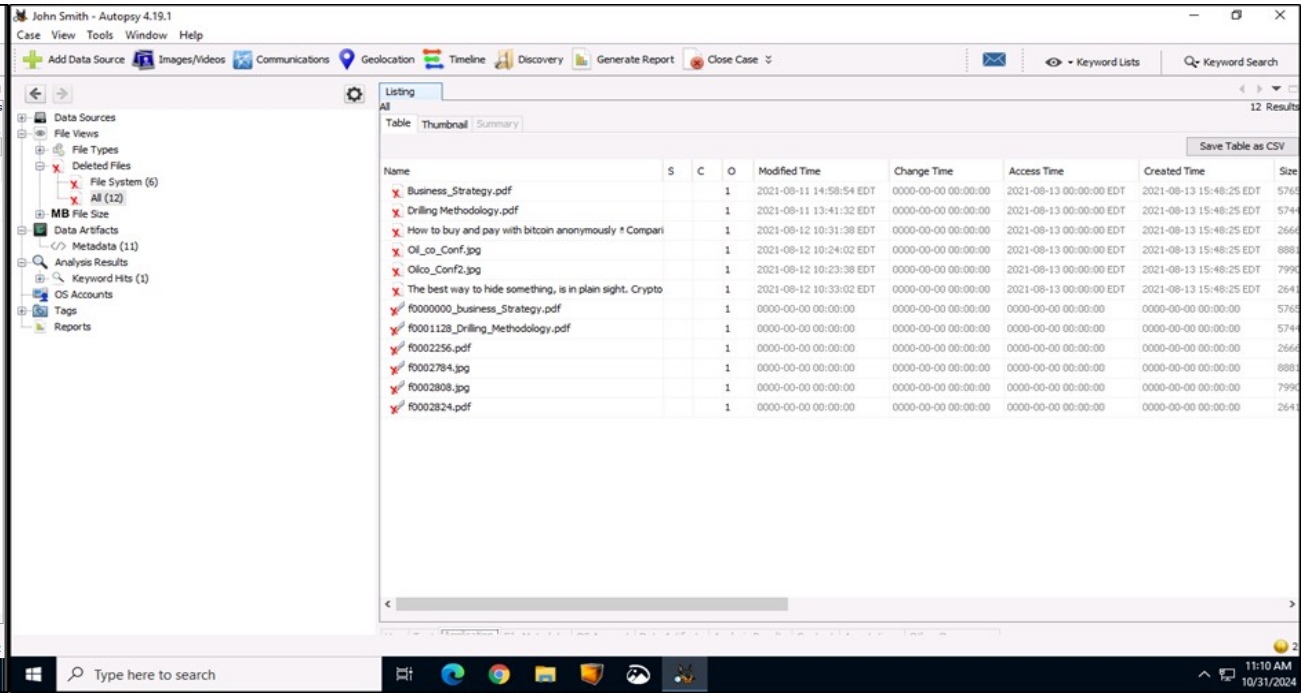




Phase 2: Deleted File Analysis

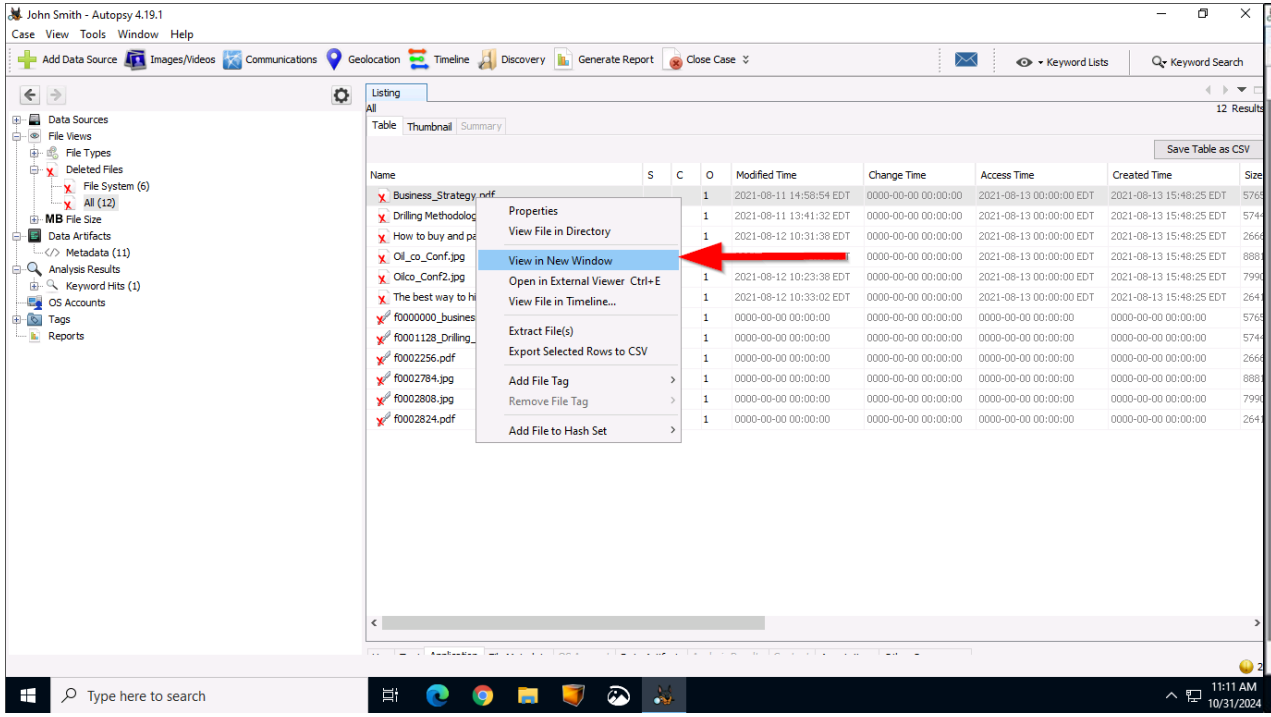
Deleted file artifacts were examined to search for relevant evidence. Special attention was given to recovered PDF documents, images, and archived content that may indicate unauthorized access to IP.



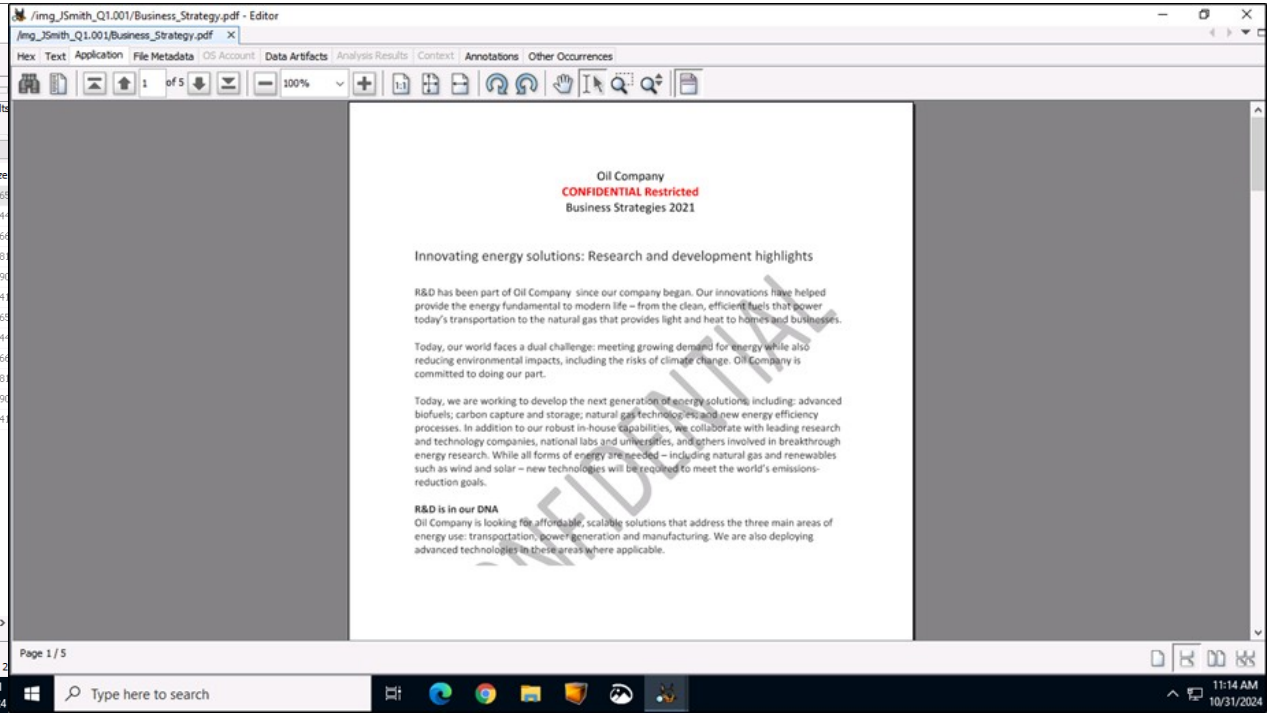


We will examine each file from
top to bottom, beginning with “Business_Strategy.pdf”

Examining Deleted Files

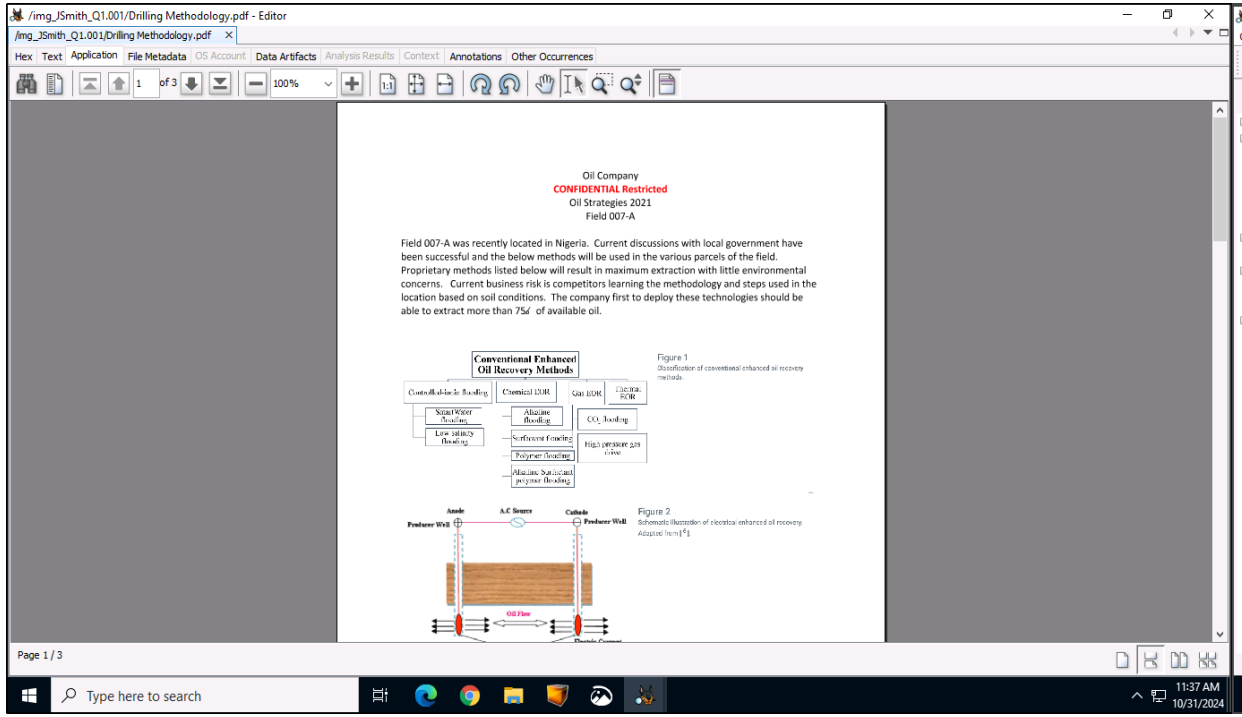


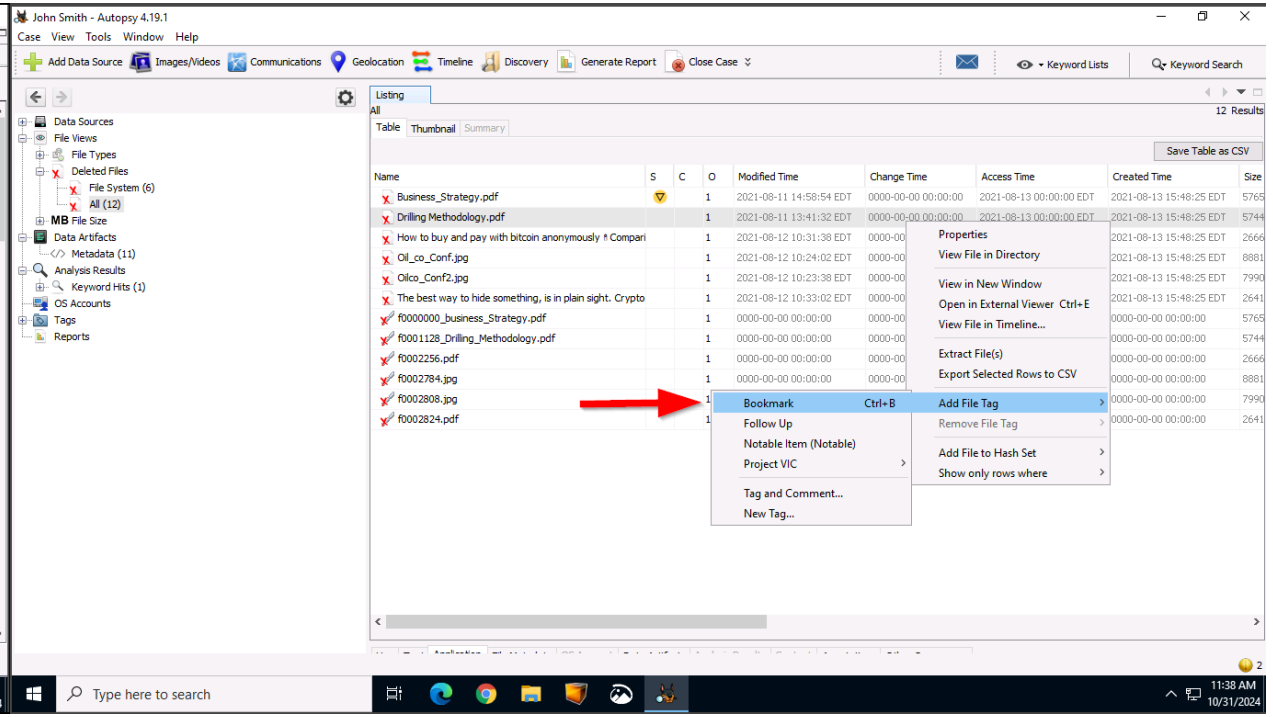
Opening “Business_Strategy.pdf” and selecting “View in New Window”



“Business_Strategy.pdf” is marked “CONFIDENTIAL Restricted”. I call this Intellectual Property Evidence One (IPE1).

Examining Deleted Files

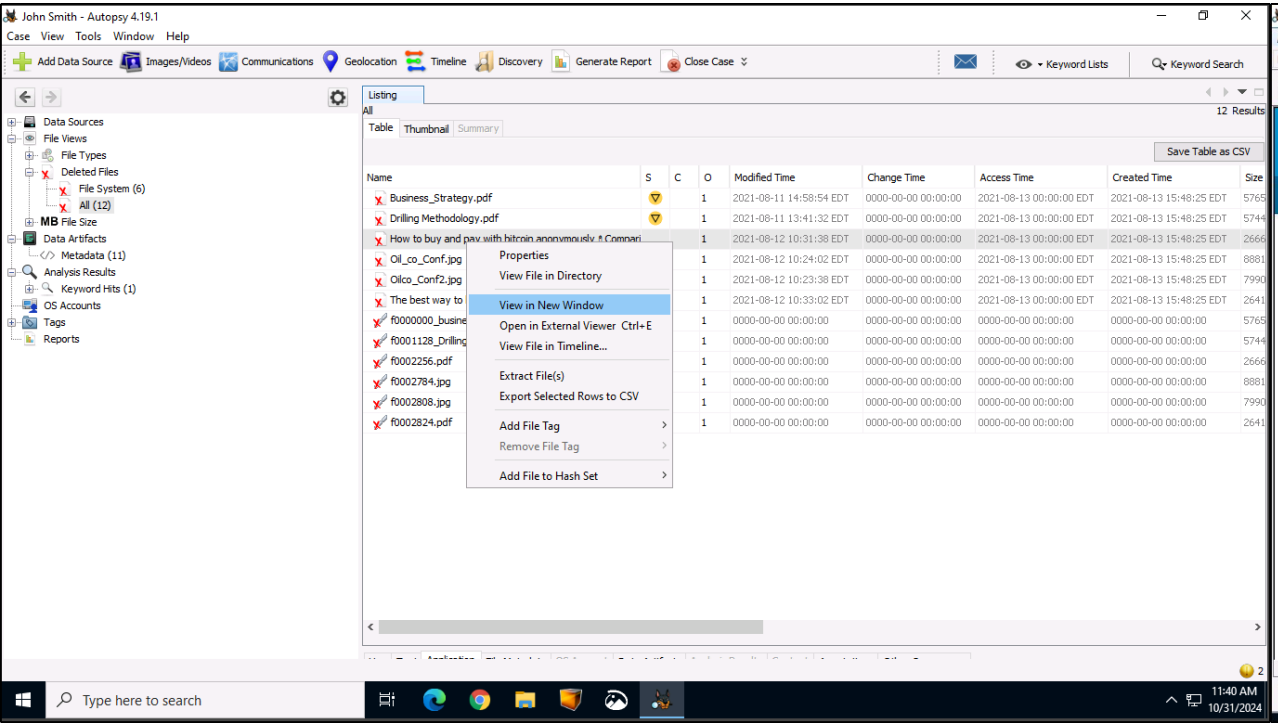


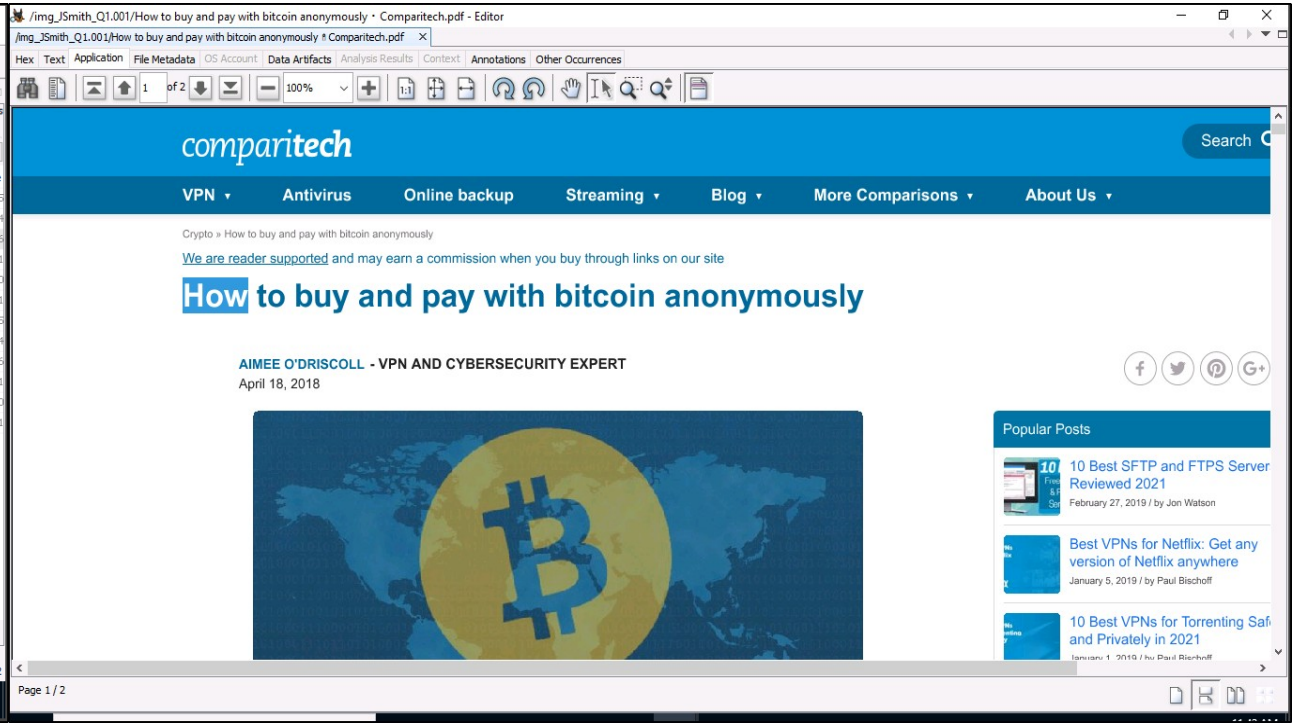


"Drilling Methodology.pdf is also marked "CONFIDENTIAL Restricted", IPE2.

Placing Bookmark for "Drilling Methodology.pdf" before moving down the list.

Examining Deleted Files

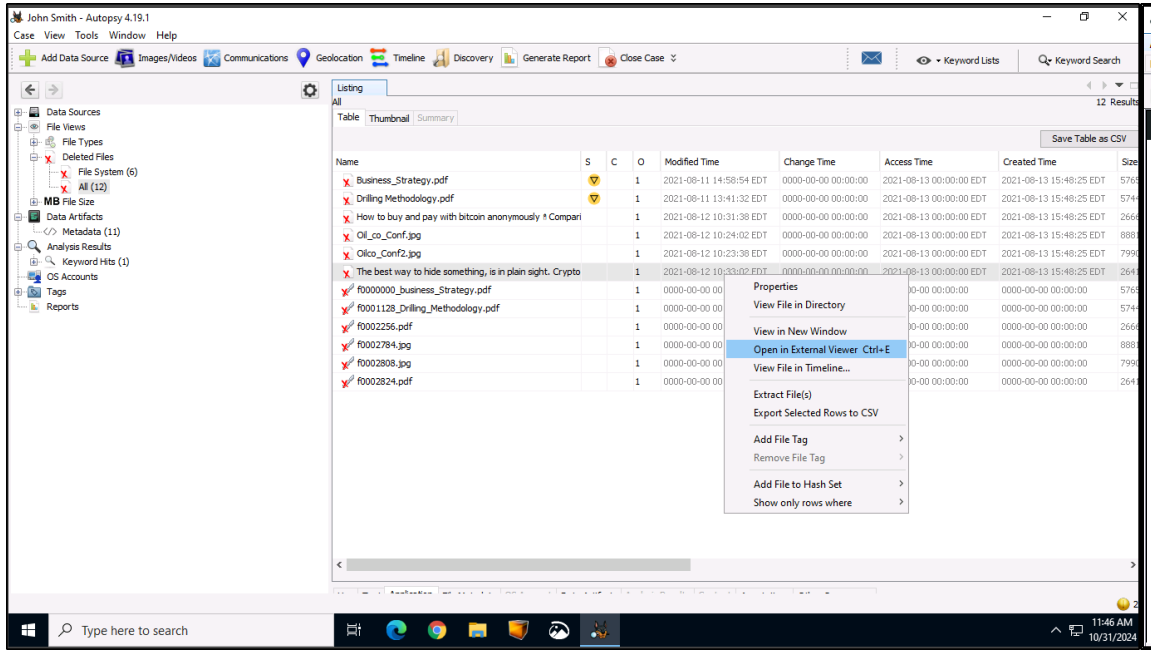


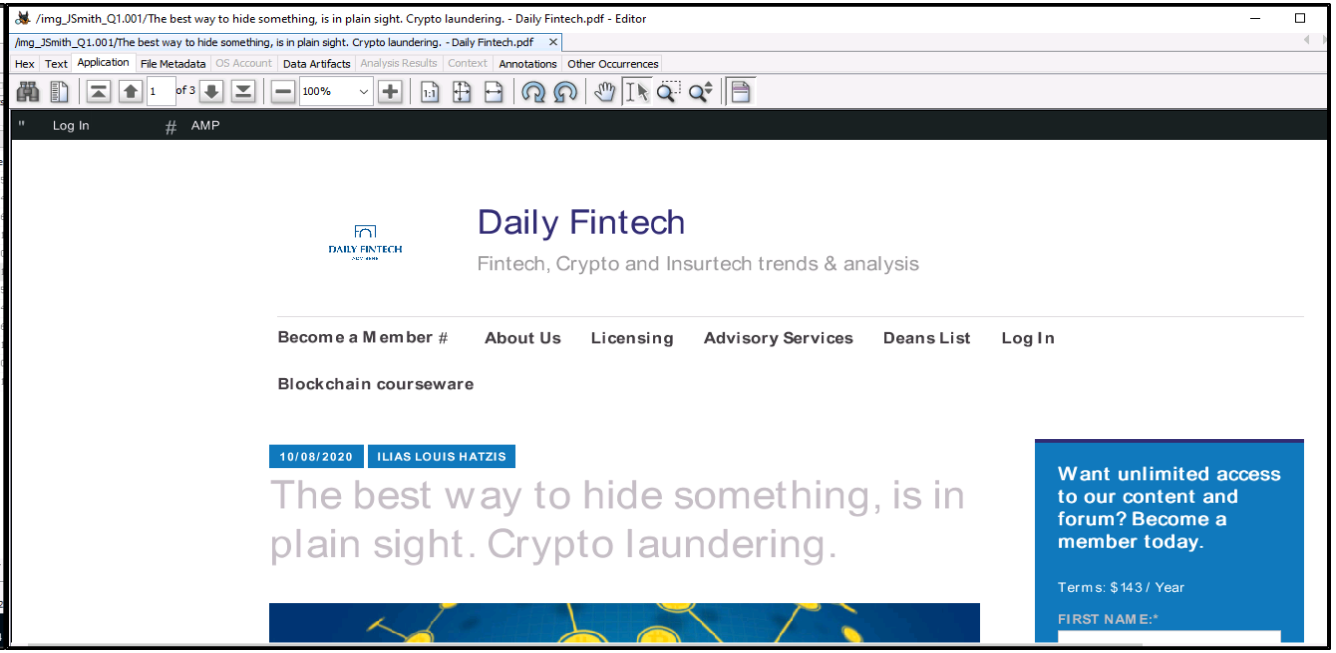


Right Clicking on “How to buy and pay with bitcoin anonymously Comparitech.pdf” to View in New Window

“How to buy and pay with bitcoin anonymously Comparitech.pdf” appears to be an article related to anonymous bitcoin transactions. Perhaps suspicious, but not directly connected with scope of investigation (stolen IP)

Examining Deleted Files

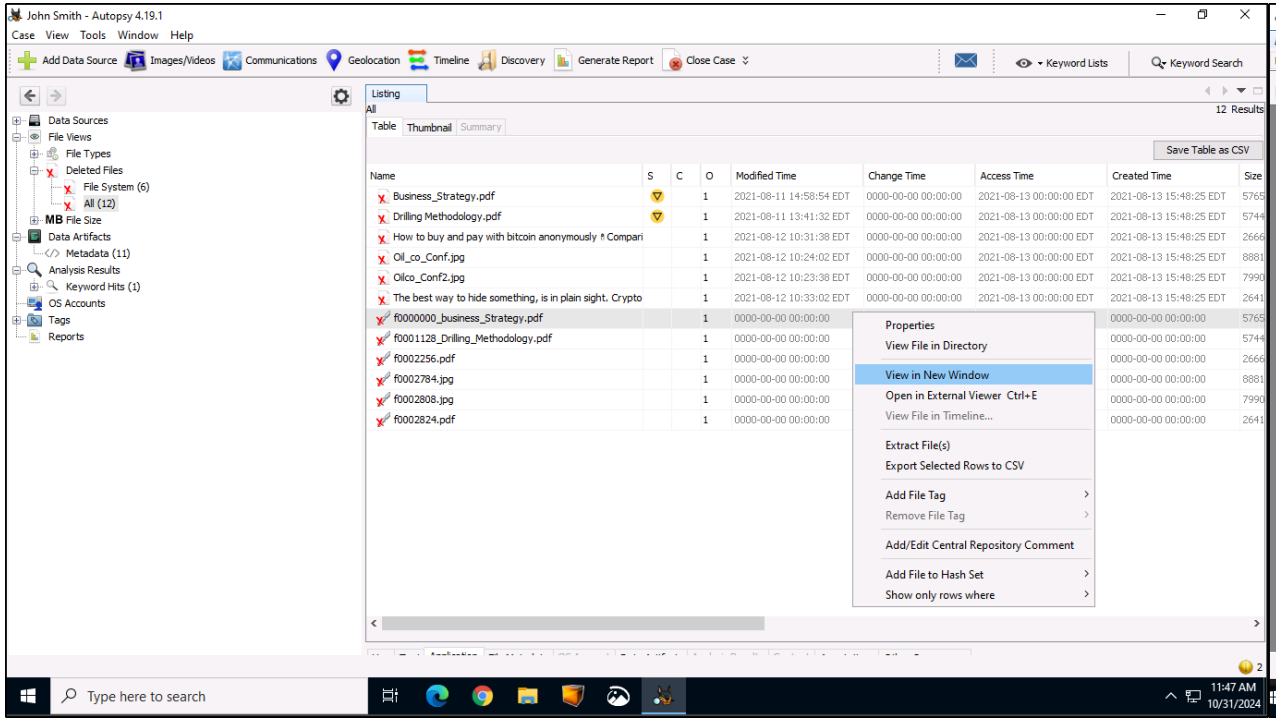


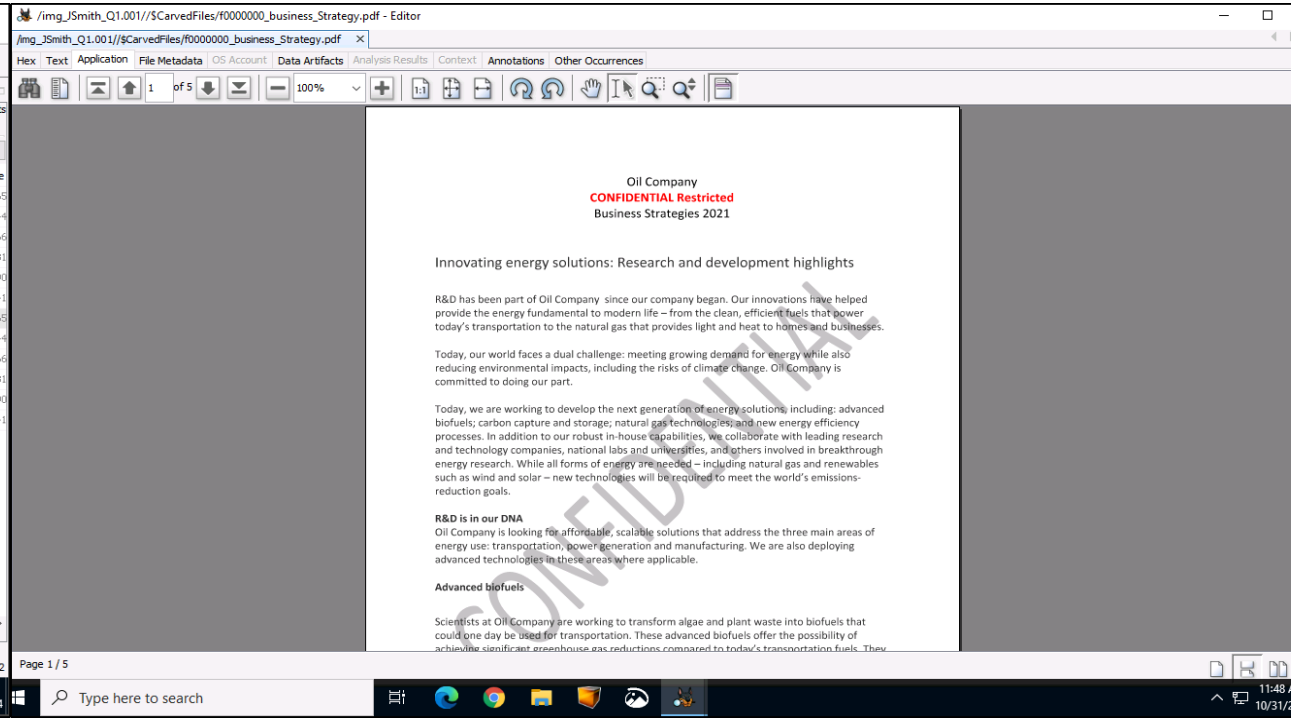


Right Clicking on “The best way to hide something, is in plain sight. Crypto laundering. – Daily Fintech.pdf” to View in New Window.

“The best way to hide something, is in plain sight. Crypto laundering. – Daily Fintech.pdf” appears to be another article on suspicious activities, but not directly tied to IP theft.

Examining Deleted Files

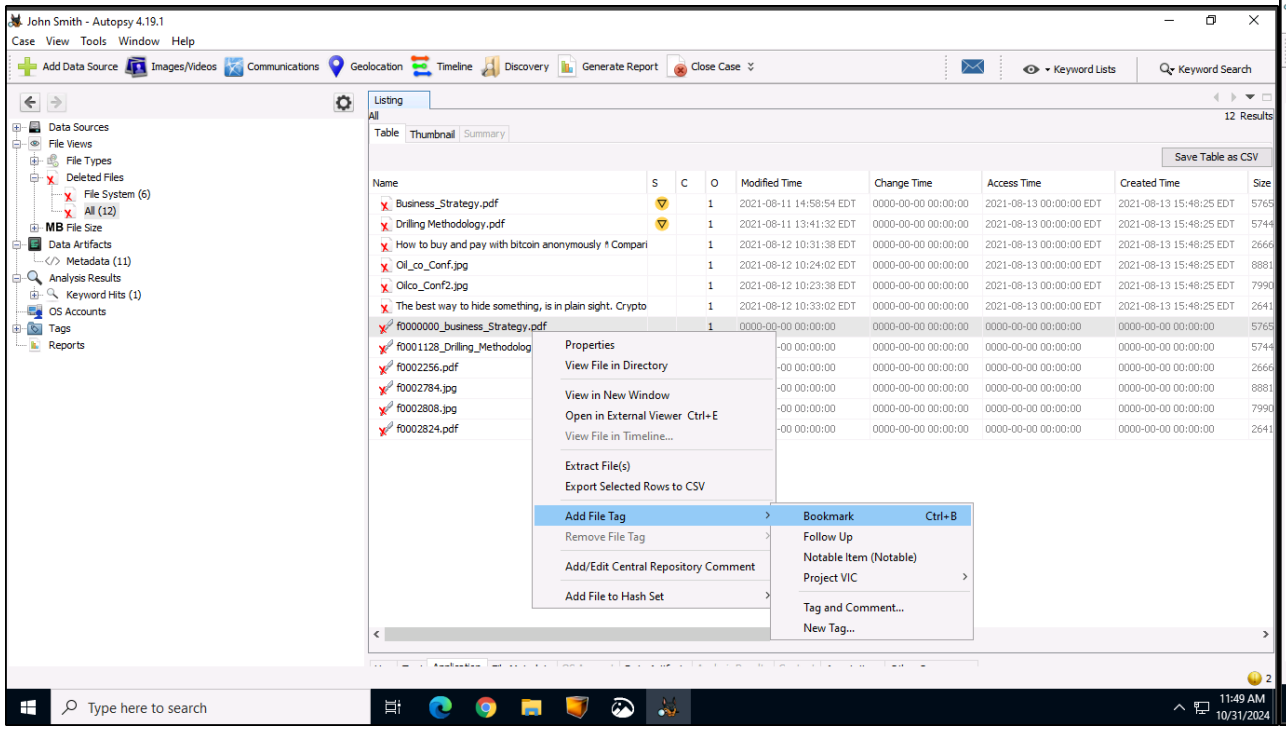




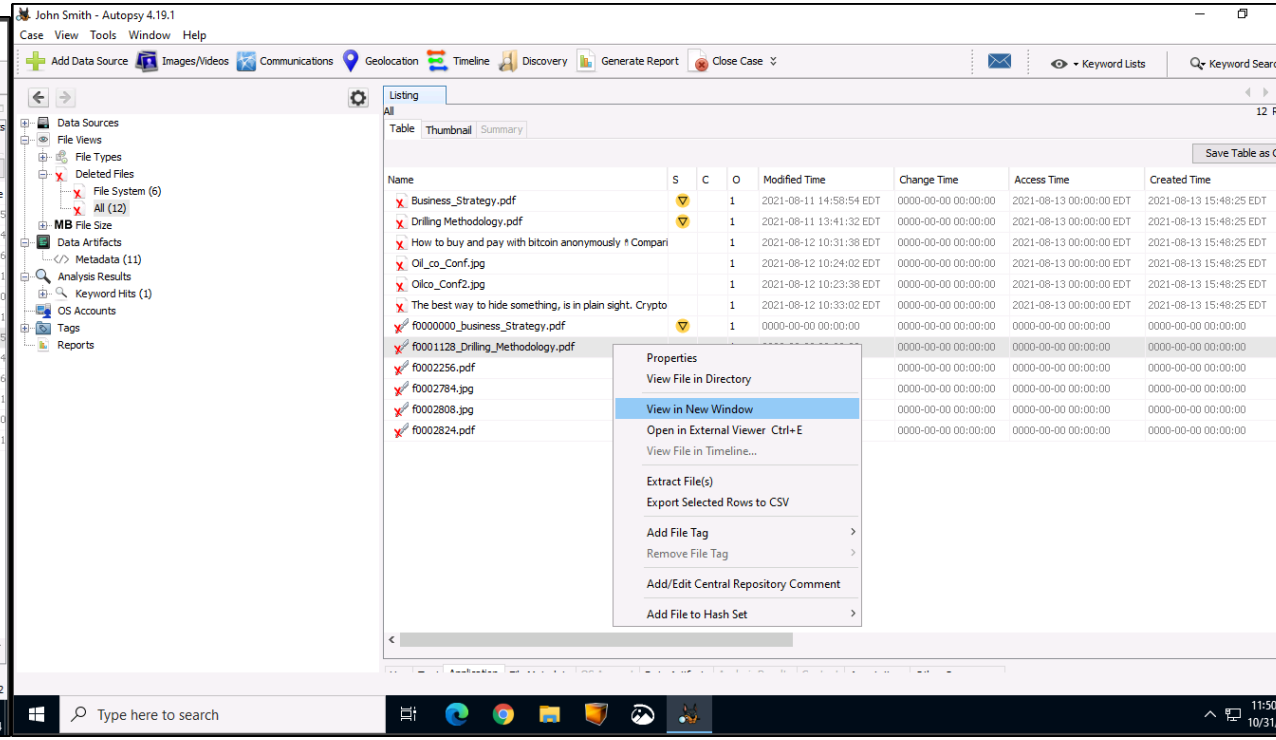
Right Clicking on "f0000000_business_Strategy.pdf" to View in New Window.

"f0000000_business_Strategy.pdf" is clearly labeled "CONFIDENTIAL Restricted", IPE3.

Examining Deleted Files

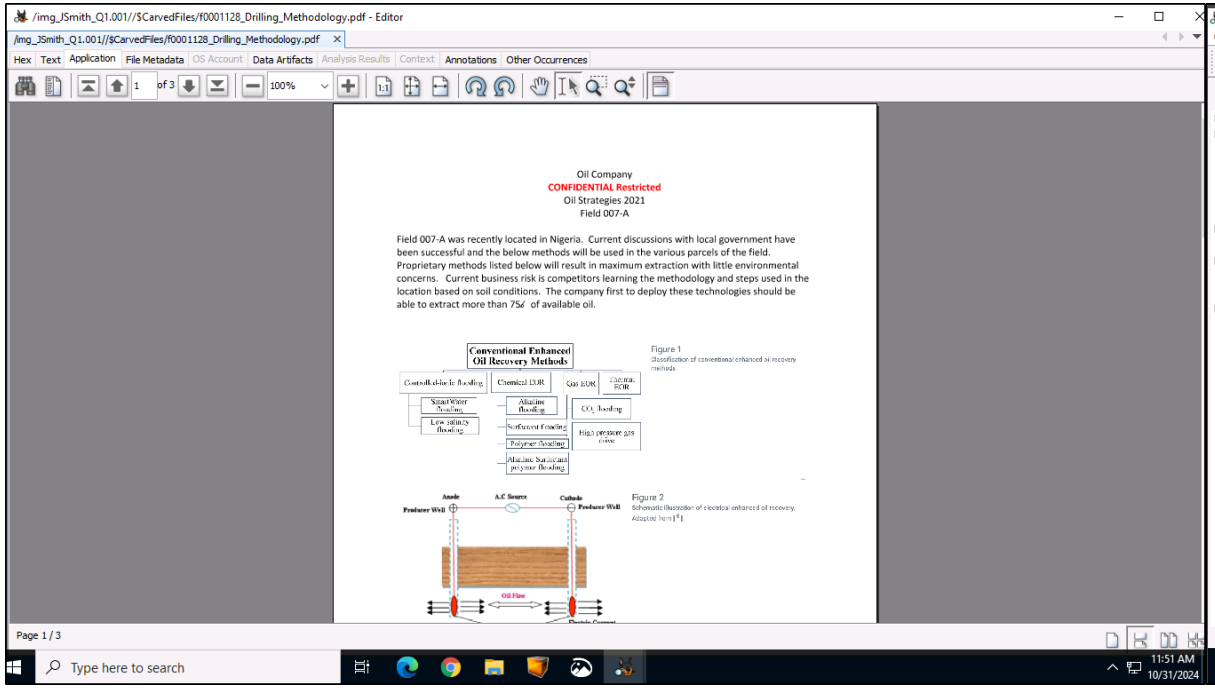


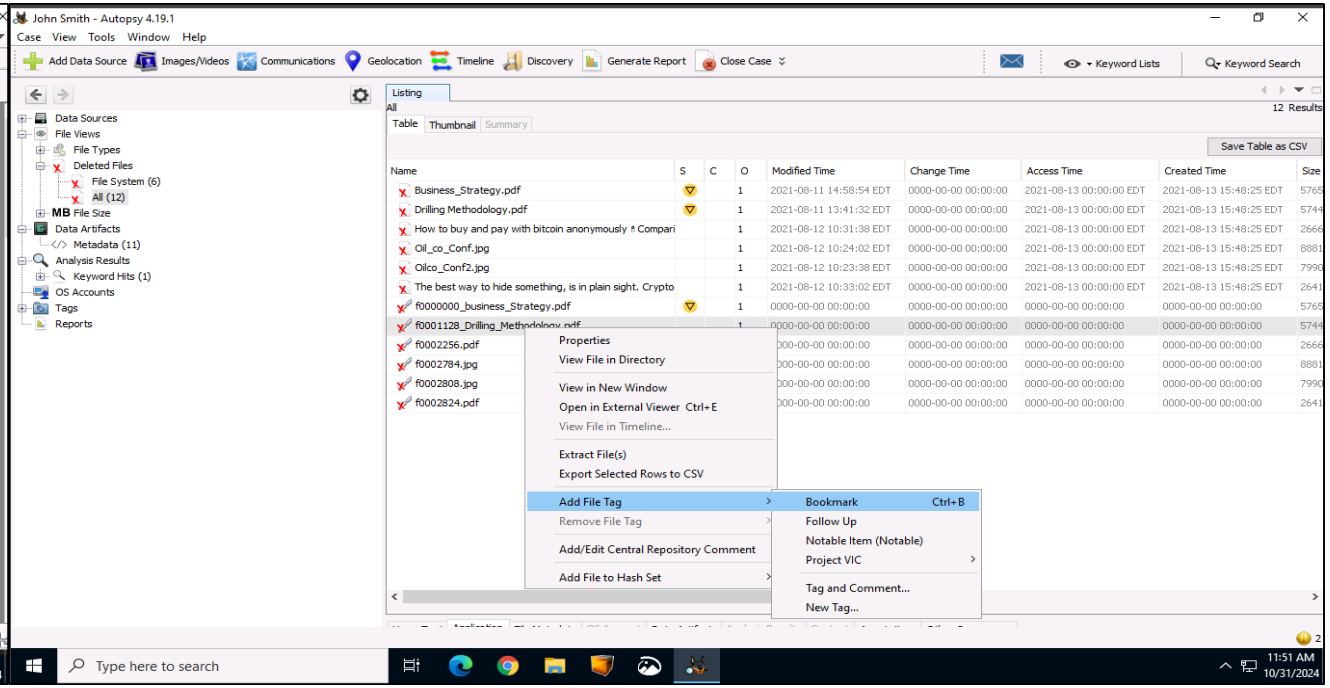
Placing Bookmark for “f0000000_business_Strategy.pdf” before moving down the list.



Right Clicking on “f0001128_Drilling Methodology.pdf” t
View in New Window.

Examining Deleted Files

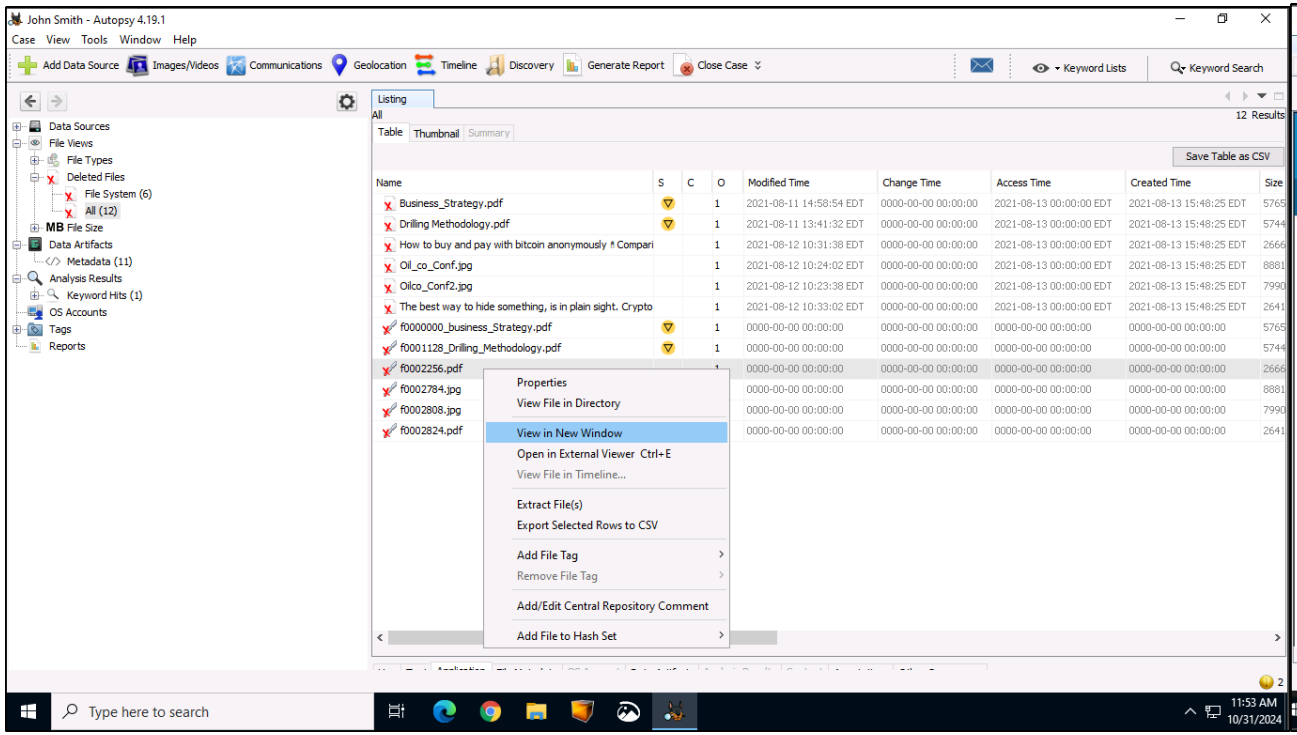


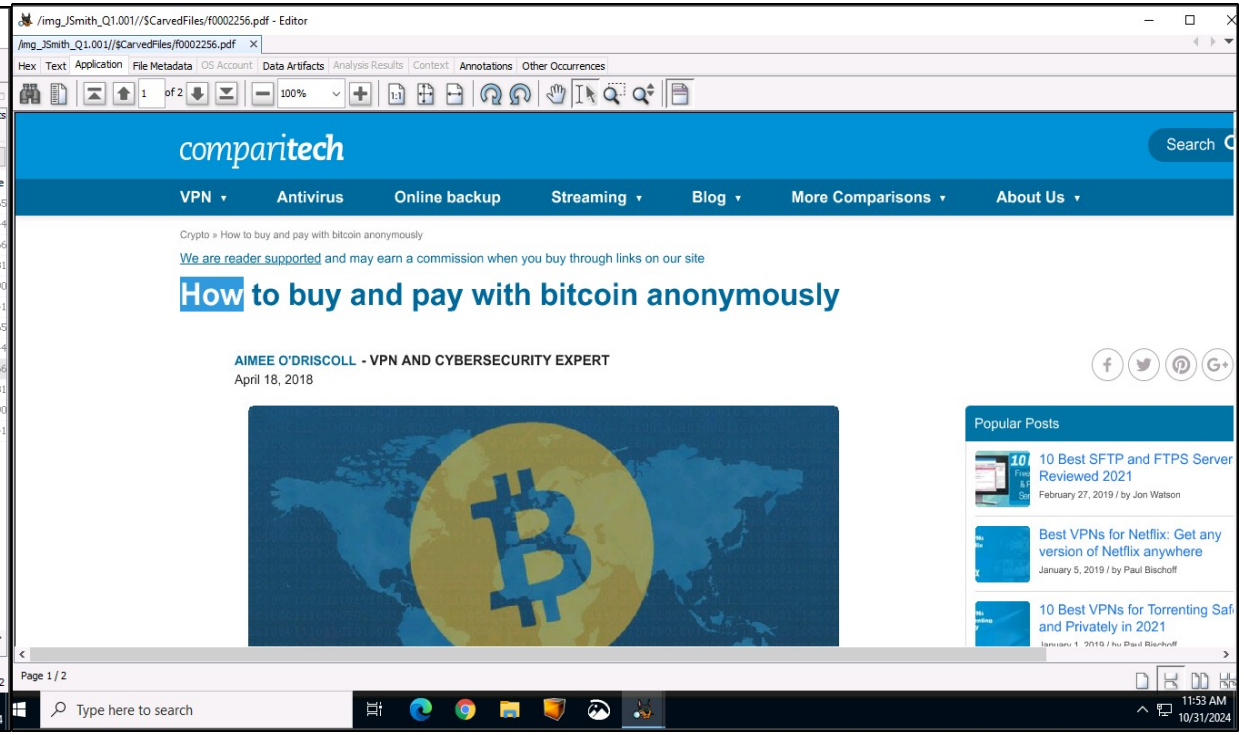


"f0001128_Drilling Methodology.pdf" is clearly labeled "CONFIDENTIAL Restricted", IPE4.

Placing Bookmark for "f0001128_Drilling Methodology.pdf" before moving down the list.

Examining Deleted Files

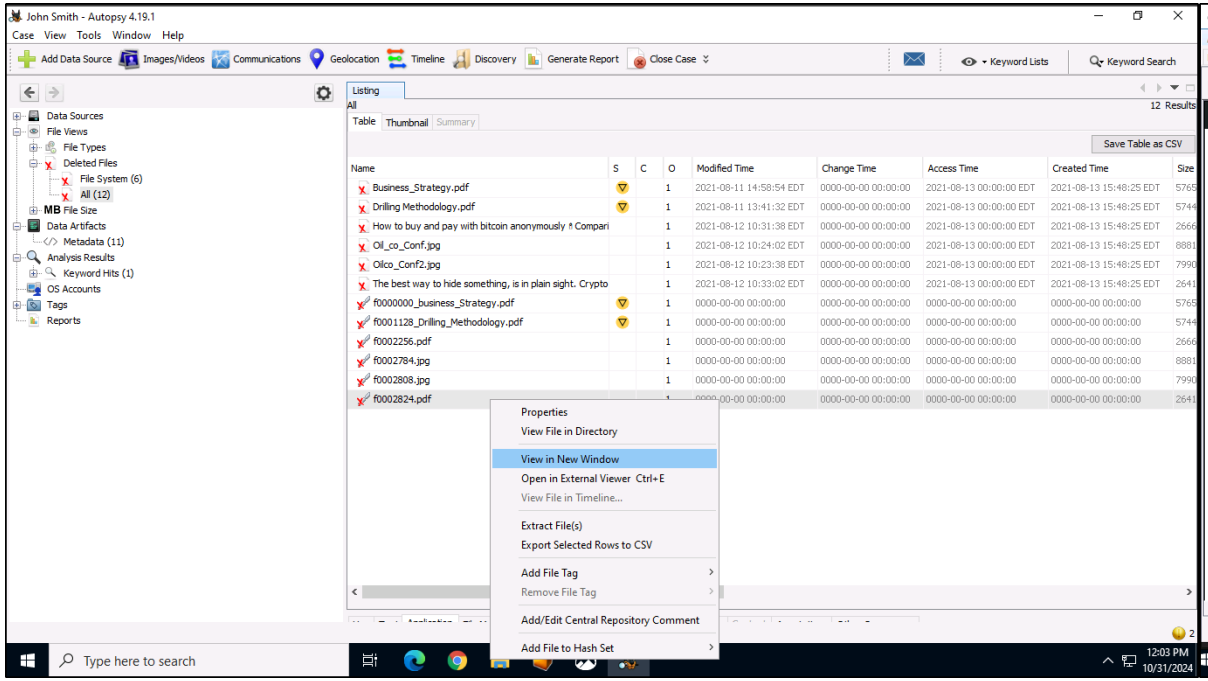


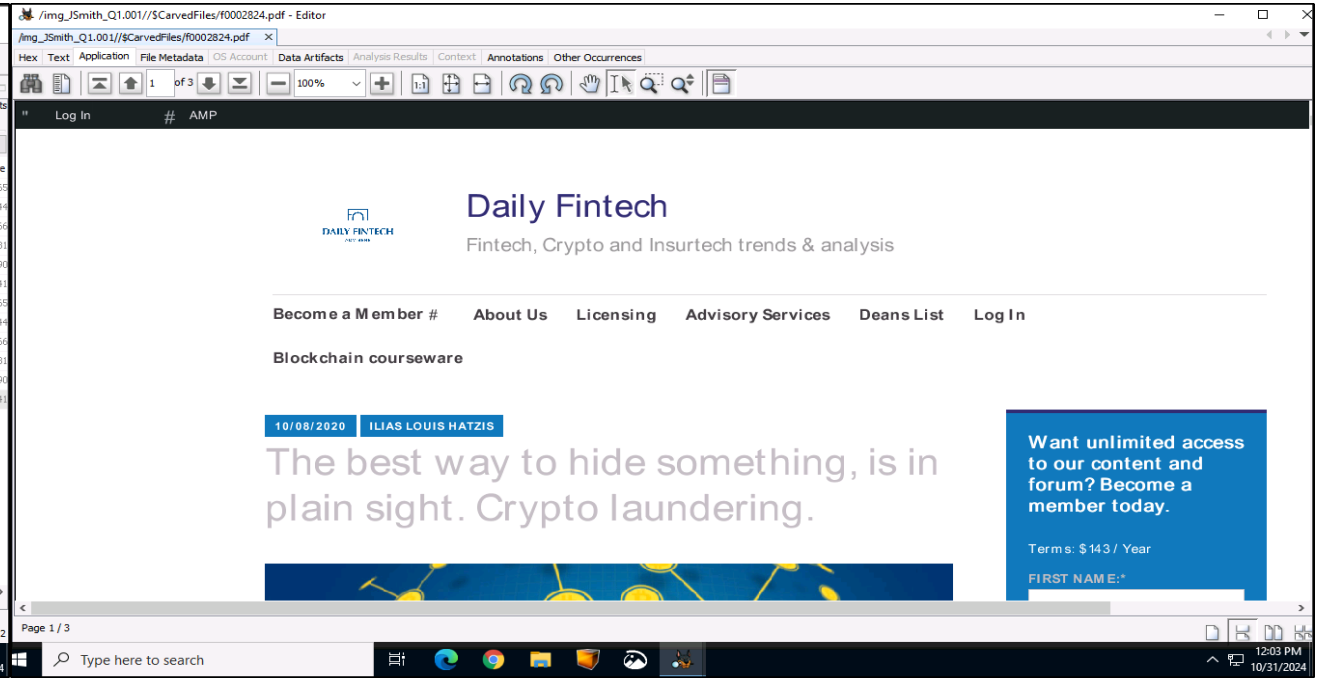


Right Clicking on "f0002256.pdf" to View in New Window.

"f0002256.pdf" appears to be another article on suspicious activities, but not directly tied to I.P. theft.

Examining Deleted Files

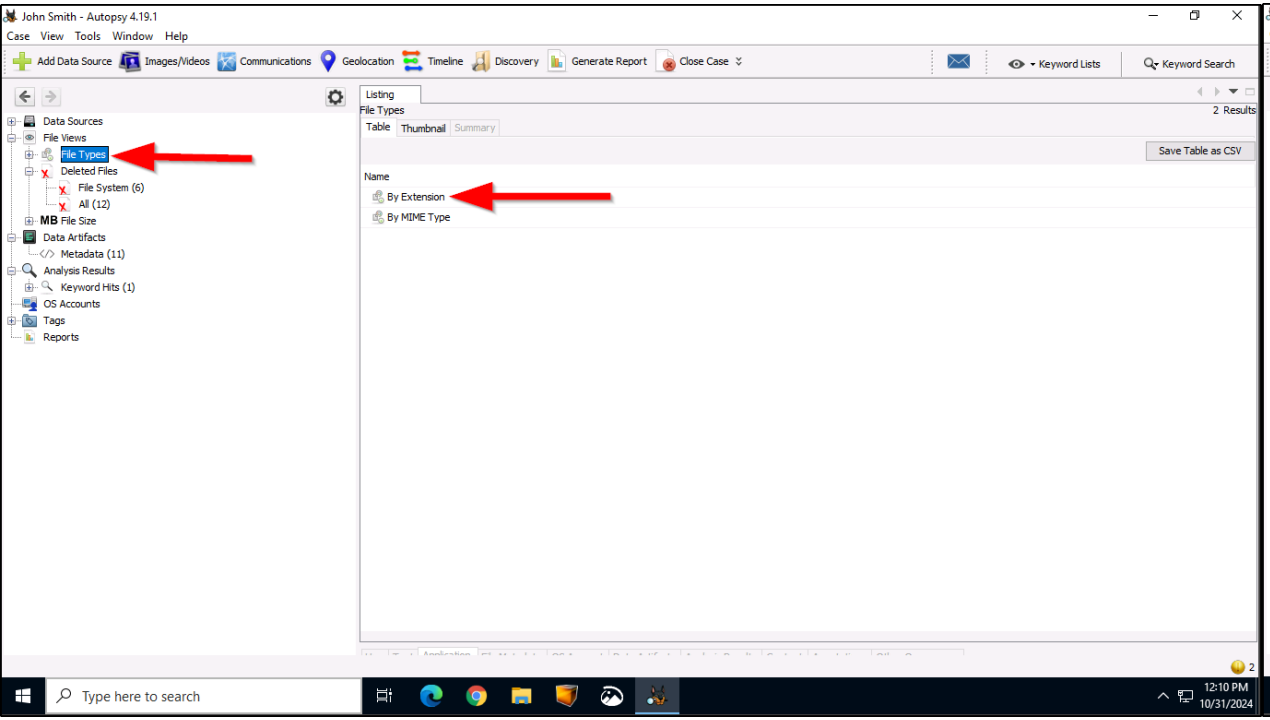


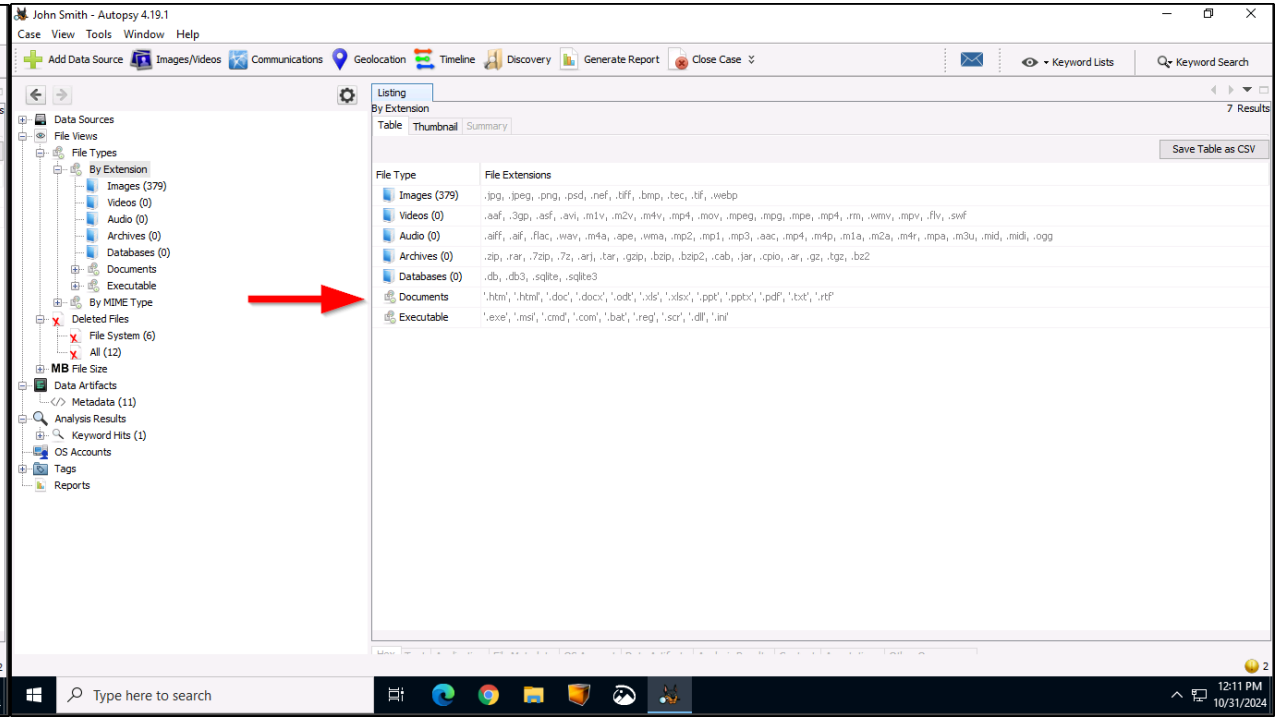


Right Clicking on "f0002824.pdf" to View in New Window.

"f0002824.pdf" appears to be another article on suspicious activities, but not directly tied to IP theft.

Examining Deleted Files

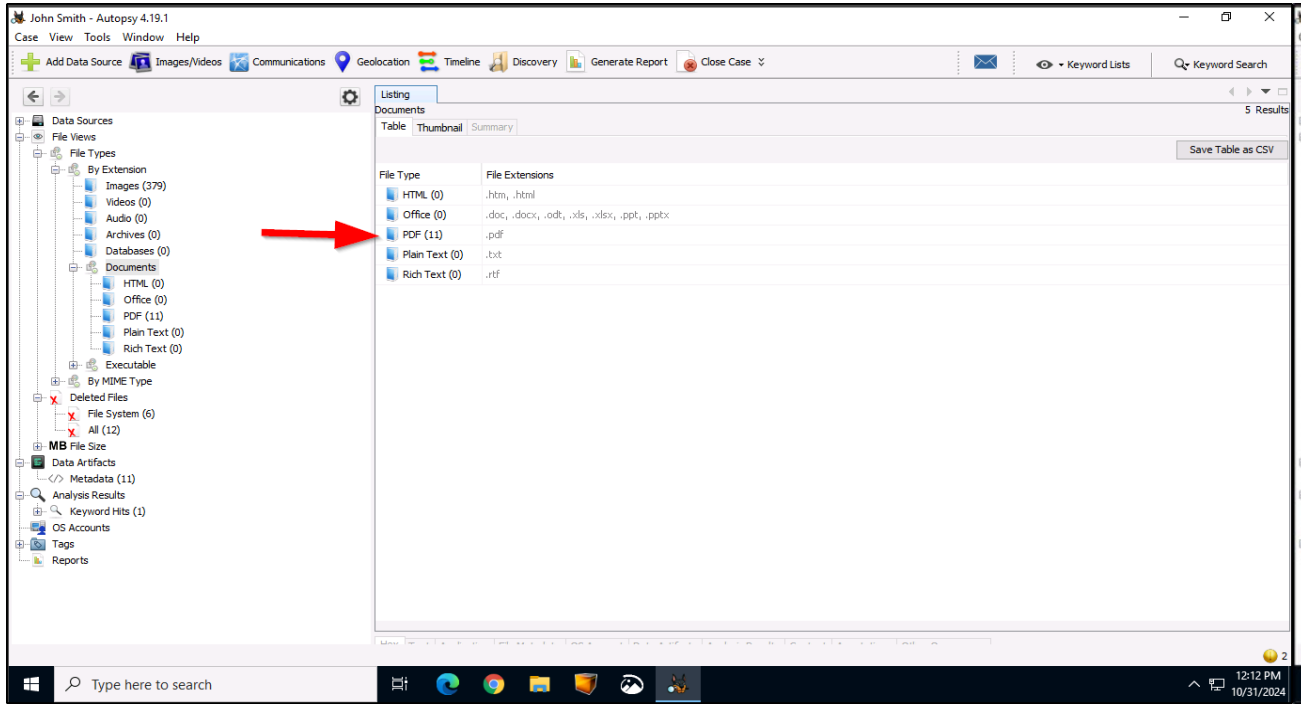




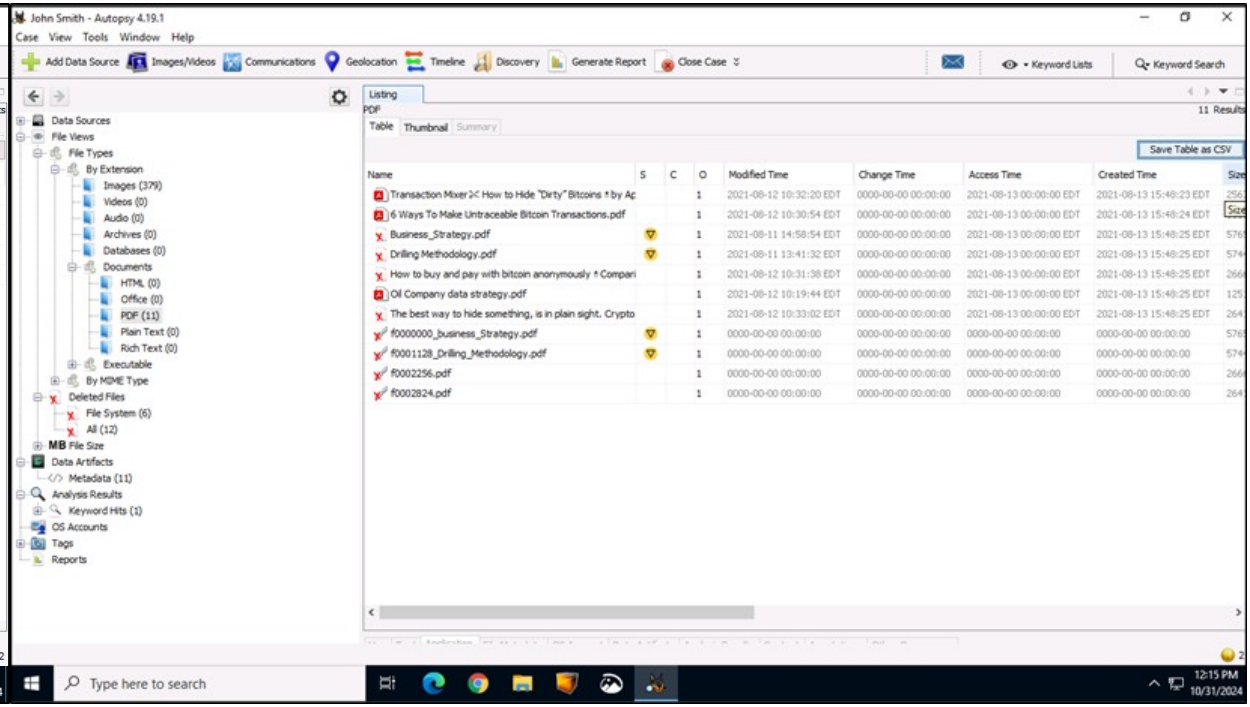
The search of "Deleted Files" uncovered 4 PDF files that were clearly labeled as IP; I think it would be wise to look at all PDF located on the evidence drive. I selected "File Types" > "By Extension".

I then selected "Documents".

Examining Deleted Files

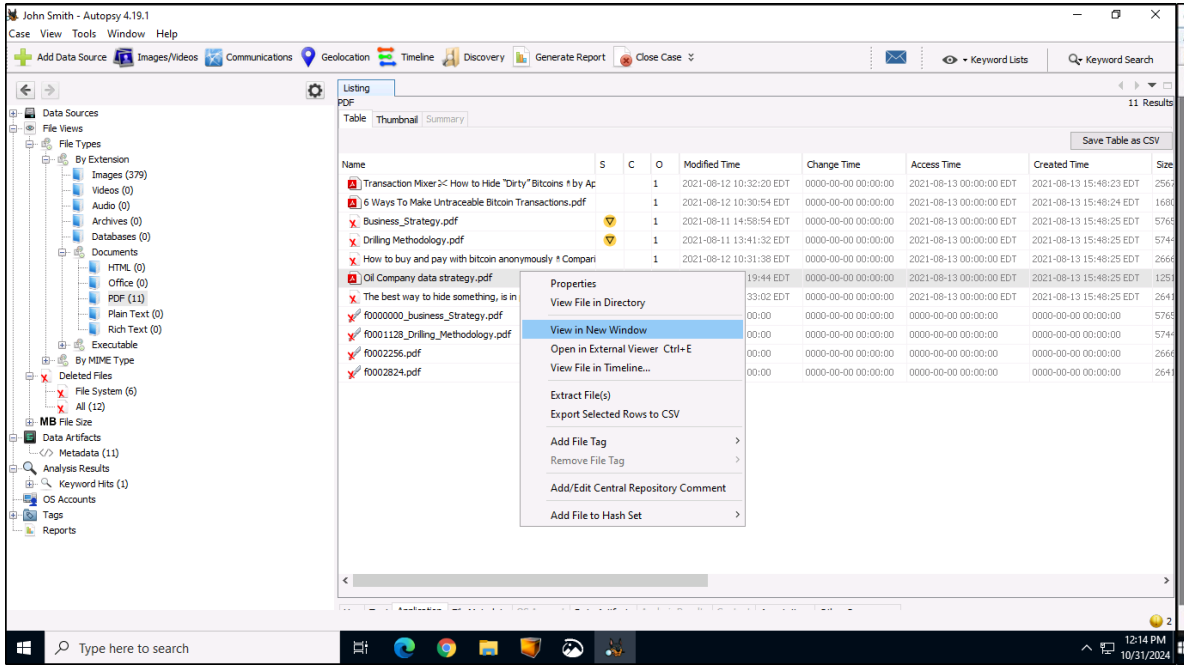


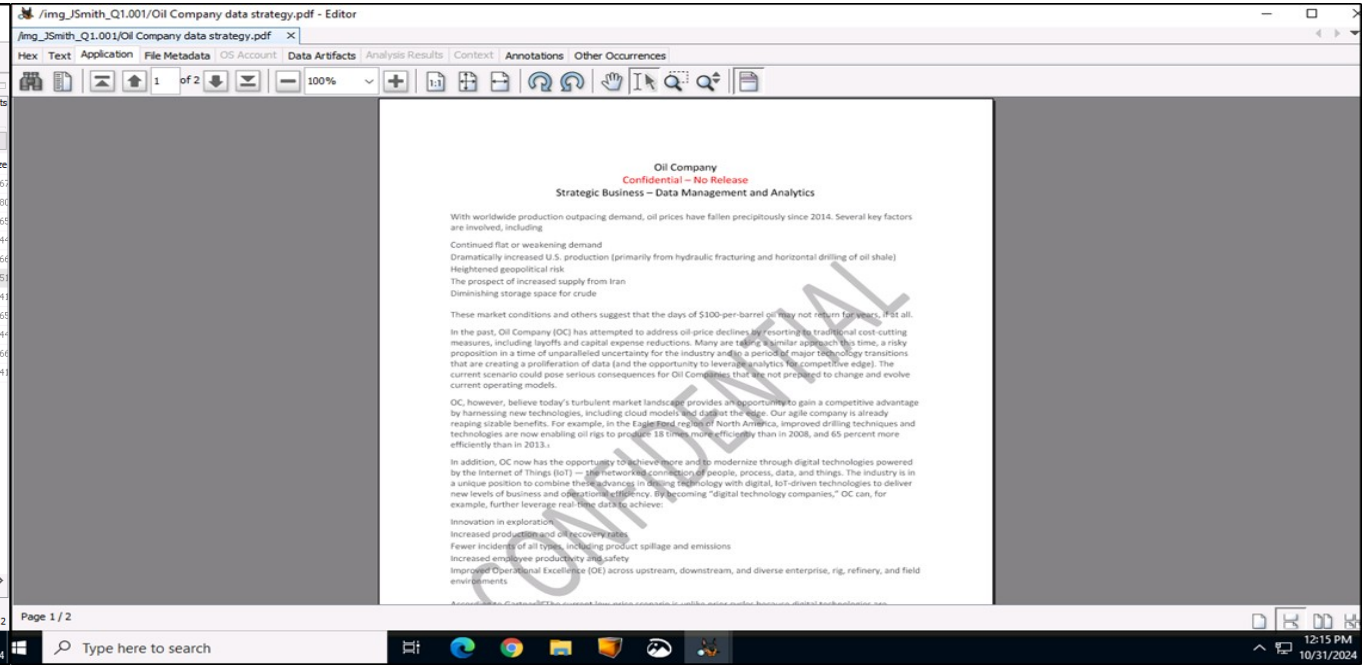
I then selected the “PDF” category.



As we see, this list already contains the four IPEs I have Bookmarked previously.

Examining Deleted Files

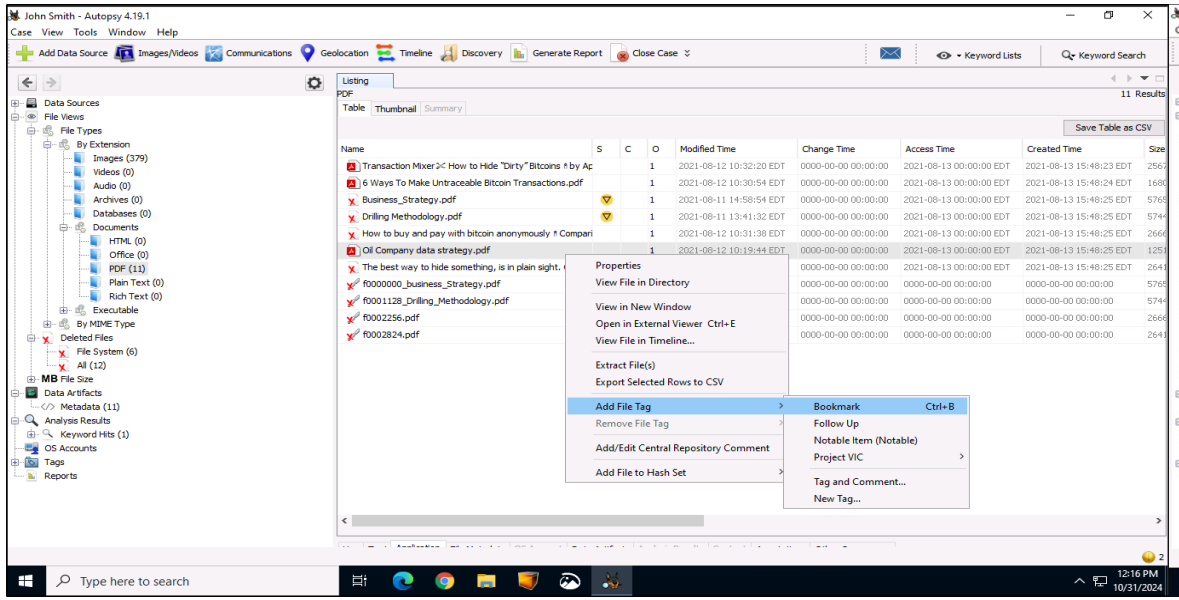




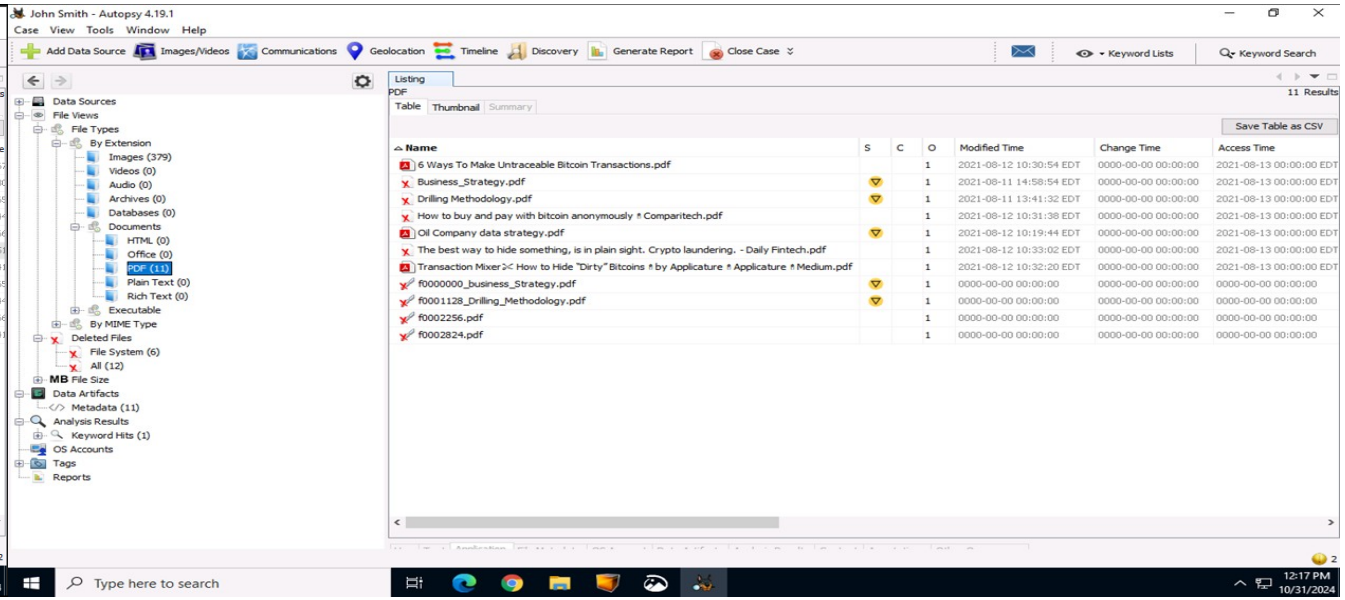
Right Clicking on "Oil Company data strategy.pdf" to View in New Window.

"Oil Company data strategy.pdf" is clearly marked "Confidential – No Release", IPE5

Examining Deleted Files



Placing Bookmark for
"Oil Company data strategy.pdf"



Screenshot showing the 5 instances of IPE PDF files found
On suspect's device.

Finding 1: Confidential Business Strategy Documents

Multiple recovered files were clearly labeled as confidential business strategy documents.

- Evidence identified:
  - Business_Strategy.pdf
  - f0000000_business_Strategy.pdf

Both artifacts contained visible confidentiality markings.

Finding 2: Proprietary Engineering Documentation

The investigation uncovered confidential engineering documentation relating to drilling methodologies and operational processes.

- Evidence identified:
  - Drilling Methodology.pdf
  - f0001128_Drilling Methodology.pdf
- Both documents were marked "CONFIDENTIAL Restricted."

Finding 3: Confidential Corporate Strategy Documentation

- An additional confidential document titled “Oil Company Data Strategy.pdf” was recovered. The file was marked “Confidential – no release”.

Evidence Summary

Evidence ID	File Name	Description	Classification
IPE-1	Business_Strategy.pdf	Business Strategy Document	Confidential - Restricted
IPE-2	Drilling Methodology.pdf	Drilling Methodology Document	Confidential - Restricted
IPE-3	f0000000_business_Strategy.pdf	Recovered Business Strategy Artifact	Confidential - Restricted
IPE-4	f0001128_Drilling Methodology.pdf	Recovered Drilling Methodology Artifact	Confidential - Restricted
IPE-5	Oil Company data strategy.pdf	Oil Company Data Strategy Document	Confidential – No Release

Conclusion

- Fundamental Question: Is there any digital evidence that Mr. Smithers accessed said IP?
- Answer: Yes, the forensic examination identified five artifacts containing confidential corporate information, including business strategy and engineering-related documents. Several artifacts were recovered from deleted-file locations, supporting the allegation that protected intellectual property is stored on the device. The findings justify additional review by legal, human resources, and security personnel.
- Potential Business Impact: Theft of proprietary business strategies, exposure of engineering methodologies, competitive disadvantage, and loss of intellectual property